BEIXI (CHLOE) DU

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EDUCATION

Cornell University, College of Engineering Expected Dec 2026
Master of Engineering in Financial Engineering | GPA: 3.951 *Ithaca, NY*

- Selected coursework: Applied Time Series; Financial Engineering with Stochastic Calculus; Optimization; Fixed-Income Securities and Interest-Rate Options; Monte Carlo Simulation; Big Messy Data; Operations Research Tools for Financial Engineering.

McGill University May 2025
Bachelor of Arts in Mathematics, Minor in Computer Science and Economics | GPA: 3.96/4.00 *Montreal, Canada*

- Selected coursework: Applied Machine Learning; Advanced Calculus; Real Analysis; Ordinary Differential Equations; Numerical Analysis; Algorithms and Data Structures; Applied Linear Regression; Stochastic Processes.
- Honors: Dean's Honour List; McGill Alumnae Susan Cameron Vaughan Scholarship.

MANUSCRIPTS

Dual Control of Linear Systems from Bilinear Observations with Belief Space Model Predictive Control
Daniel Cao*, **Beixi Du***, Andrew Lowitt*, Sunmook Choi*, Sarah Dean, Yahya Sattar
Submitted to CDC 2026; ****equal contribution***
Studies finite-horizon quadratic control of linear systems with bilinear observations, where control inputs affect both system dynamics and observation quality. Proposes belief-space model predictive control (B-MPC), which plans over both state estimates and input-dependent estimation covariance.

The Prestige: Benchmarking Cognitive Visual Reasoning using Magic Tricks
Shuo Wen, **Beixi Du**, Junming Shi, Edwin, Chuqiao Si
Submitted to NeurIPS 2026
Introduces a video-understanding benchmark of professional magic performances for evaluating VLM cognitive visual reasoning. Contains 180 videos and 2,177 open-ended questions across contextual, paradoxical, instant, temporal-summarization, and adversarial false-premise categories.

RESEARCH EXPERIENCE

Reasoning-Aware Reward Models for Vision-Language-Action (VLA) Evaluation June 2026 – Present
Cornell University, Department of ORIE *Ithaca, NY*

- Proposing a human-preference–trained, *reasoning-aware* reward model that jointly scores a VLA policy's reasoning trace and action trajectory as a cheap, label-efficient automatic evaluator, targeting the statistically underpowered and human-labor-intensive real-robot evaluation problem.
- Designing a GenRM-style multimodal reward model (LoRA-adapted open VLM backbone, Qwen2-VL) that emits a critique then a Bradley–Terry preference, with Math-Shepherd–style Monte-Carlo rollout value estimates amortized by a learned value model for dense per-step credit assignment.
- Built an end-to-end evaluation harness (PyTorch / HF Transformers) with paired video-only vs. reasoning-aware judging and Wilson-interval / two-proportion significance testing; designed a controlled reasoning-ablation protocol over ECoT + SimplerEnv rollouts to test reasoning–action faithfulness against reasoning-blind baselines (Robometer, VLP).

LLM Agent Policy Recovery from Noisy Human Demonstrations Sep 2025 – Present
Cornell University, Department of Computer Science | Advisor: Prof. Kilian Q. Weinberger *Ithaca, NY*

- Developed a two-stage pipeline for recovering high-quality LLM agent policies from noisy human demonstrations by combining graph-based policy extraction with cross-trajectory statistical gap detection.

- Analyzed 74 demonstration trajectories to identify systematic workflow shortcuts missed by bullet-point extraction baselines such as ACE.
- Conducted ablations across graph refinement, delta batch refinement, gap detection, and failure-contrast sharpening.
- Improved agent pass@1 from 67.5% under the ACE low-quality-trajectory baseline to 90.0%, approaching the high-quality demonstrator ceiling of 95.0%; selective sharpening increased pass@3 consistency from 42.5% to 55.0%.

Belief-Space Control for Bilinear Observation Systems Sep 2025 – Apr 2026
Cornell University, Department of Computer Science | Advisor: Prof. Sarah Dean *Ithaca, NY*

- Contributed equally to a CDC 2026 submission on dual control of linear dynamical systems with bilinear observations, where the separation principle fails because control inputs affect future observation quality.
- Implemented and benchmarked three controllers—B-MPC, Sep, and Sep-MPC—across two synthetic systems: a random bilinear observation system and a multi-block double-integrator system.
- Built the PyTorch experimental pipeline with L-BFGS optimization; analyzed cost-vs-horizon sweeps, Kalman-filter diagnostics, counterfactual action scatter plots, and uncertainty-vs-action-gap behavior.
- Showed that B-MPC reduces control cost by explicitly planning over both the belief mean and error covariance under an input-dependent Kalman filter.

Quantitative Research Intern Jul 2024 – Oct 2024
Western Securities Research & Development Center *Shanghai, China*

- Built a time-series linear regression model relating real interest rates and the equity risk premium ($R^2 = 0.363$), identifying a statistically significant negative correlation for asset-allocation analysis.
- Designed an NLP pipeline using BERT to extract and quantify individual-stock sentiment from financial news, investor forums, and analyst reports in the China A-share market.
- Enhanced the sentiment signal with customized time-decay weighting schemes for downstream financial analysis.

Analyst Intern May 2023 – Jul 2023
Huatai Securities Co., Ltd. *Shanghai, China*

- Modeled daily returns and volatility using ARIMA and GARCH models; identified reversal effects in Innovent Biologics and SSY Group Ltd.
- Conducted correlation analysis against the Hang Seng Health Care Index and Hang Seng Index.
- Supported pharmaceutical-sector research by integrating policy analysis, macroeconomic indicators, and firm-level financials from 2018–2023.
- Applied DCF valuation using WACC assumptions to identify potentially undervalued companies for buy recommendations.

TECHNICAL SKILLS

Programming: Python, PyTorch, NumPy, R, Java, C, MATLAB, Bash, Linux
ML / Statistics: Applied machine learning, time-series modeling, stochastic processes, Monte Carlo simulation, optimization, PCA, regression, ARIMA, GARCH
Finance Tools: Bloomberg, Excel, fixed-income analytics, financial engineering, risk and valuation modeling
Certifications: Bloomberg Market Concepts; AI Agents Fundamentals; CFA Level I Candidate

LEADERSHIP

Team Member, Activity Planning Department Sep 2022 – May 2023
McGill University *Montreal, Canada*

- Planned and hosted a psychology night talk, “How to Love Ourselves,” attended by 30 students.
- Collaborated with team members to organize two team-building events.